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Metabolic Alterations Associated With Rapidly Progressive Chronic Kidney Disease in Cats

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Keywords: biomarkers | cats | chronic kidney disease | metabolic profiling | pathophysiology | progression

ABSTRACT

Background: Chronic kidney disease (CKD) in cats often remains stable over time, but some cats experience progressive kidney dysfunction without an identifiable cause.

Objectives: Identify differences in serum and urine compounds related to tryptophan metabolism and gut-derived uremic toxins between cohorts with non-progressive and rapidly progressive CKD.

Animals: Forty-two client-owned cats diagnosed with CKD were divided into a rapid progression group ($n = 8$) and a non-progressive control group ($n = 34$).

Methods: Prospective cohort study with comparative analysis of predictors using targeted metabolomics. Rapid progression was defined as a 25% increase in serum creatinine concentration over 6 months.

Results: Serum metabolite ratios of the serotonin pathway showed promising potential for predicting rapid CKD progression in cats: L-tryptophan/5-hydroxytryptophan (area under curve [AUC]: 0.89; 95% CI: 0.78–1; $p < 0.01$; sensitivity: 85.7%; specificity: 78.8%) and serotonin/5-hydroxyindole-3-acetic acid (AUC: 0.83; 95% CI: 0.64–1; $p < 0.01$; sensitivity: 87.5%; specificity: 76.5%). Cats with rapidly progressive CKD had significantly lower baseline urinary indoxyl-sulfate (median [Interquartile range, IQR], 0.25 [0.07–0.46] vs. 0.53 [0.36–0.81]; $p = 0.04$) and p-cresyl-sulfate (median [IQR], 0.13 [0.01–0.79] vs. 0.83 [0.32–1.27]; $p = 0.04$). Logistic regression analysis identified an association between decreased urinary indoxyl-sulfate (odds ratio [OR]: 0.03; 95% CI: 0–1; $p = 0.04$) and rapidly progressive CKD, following a similar pattern after correcting for serum creatinine concentration ($p = 0.06$).

Conclusions: Metabolites of the serotonin pathway: L-tryptophan, 5-hydroxytryptophan, and 5-hydroxyindole-3-acetic acid may serve as potential candidates for further predictive validation in CKD progression in cats. Decreased uremic toxin excretion in rapidly progressive CKD may underlie disease progression.

Abbreviations: AUC, area under the curve; CKD, chronic kidney disease; FFD, fractional factorial design; IQR, inter quartile range; IRIS, International Renal Interest Society; MAO-A, monoamine oxidase A; QCS, quality Control samples; RAAS, renin-angiotensin-aldosterone system; ROC, receiver operating characteristic; SDMA, symmetric dimethylarginine; sCr, serum creatinine concentration; TMA, trimethylamine; TMAO, trimethylamine-N-oxide; UHPLC-HRMS, ultra-high-performance liquid chromatography-high-resolution mass spectrometry; UPC, urine protein to creatinine ratio; USG, urine specific gravity.

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1 | Introduction

Chronic kidney disease (CKD) is very common in domestic cats, with a prevalence reaching 30% in cats over 12 years of age [1]. In most cases, idiopathic CKD remains stable over long periods of time, but it has been reported to account for 13.6% of domestic cat mortality from the age of five onwards [2, 3]. A substantial percentage of cats eventually die from causes unrelated to their kidney disease [4]. In some cases, however CKD progresses relatively rapidly, leading to deterioration of kidney function within a few months [4–6]. Certain factors contributing to the progression of CKD, such as hyperphosphatemia, renal proteinuria, and anemia, are well-established and often can be treated accordingly [5, 6]. Nonetheless, even with adequate management of CKD and its sequelae, some cats will experience progressive kidney deterioration without identifiable underlying causes [6–8].

The absence of identifiable causes for rapid progression of CKD is likely the consequence of incomplete understanding of the pathophysiology of CKD in cats. Tubulointerstitial inflammation and fibrosis already are present in early CKD and are strongly associated with deterioration of renal function [9, 10]. However, the factors underlying this process remain relatively unknown [11, 12]. In other words, certain ongoing pathophysiological mechanisms at the cellular and metabolite level remain to be elucidated, because current CKD biomarkers in cats often fail to provide detailed insight into the specific processes occurring within the affected kidneys [13, 14]. Our previous research recently provided further information on specific metabolic alterations in CKD in cats, using a metabolomic approach [15]. Pathways involving tryptophan, carnitine, and tyrosine metabolism were the most strongly perturbed. Most pathways showed a tendency to shift towards the formation and retention of gut-derived uremic toxins [15]. In human medicine, there is growing recognition of the role played by these toxins in inducing or exacerbating tubular inflammation and fibrosis, thereby accelerating progression of the disease [16, 17]. Initial evidence suggests that such uremic toxins also may be associated with rapid disease progression in cats [18, 19].

Therefore, we aimed to quantify differences in serum and urine metabolite abundance between cohorts experiencing non-progressive and rapidly progressive CKD by use of a targeted metabolomic approach. We specifically focused on tryptophan and its metabolites, and a panel of gut-derived uremic toxins identified in previous work with the goal of evaluating metabolic shifts preceding rapidly progressive CKD [15]. Biomarkers predicting rapid deterioration in kidney function may serve as candidates for further diagnostic validation. In addition to their predictive utility, the detection of biomarkers of rapid progression could constitute a first step to determine which pathways play an underlying role in the ongoing decline in kidney function. These findings may help identify mechanisms for future therapeutic modification in cats with CKD.

2 | Materials and Methods

2.1 | Ethics Statement

The study protocol was approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering (EC2020-081). All specimens from cats in the study were obtained prospectively from privately owned domestic cats using a proactive recruitment process. Written informed consent was obtained before inclusion in the study.

2.2 | Study Design

The target population was cats diagnosed with CKD, subsequently divided based on their progression status. The source population of this prospective cohort study consisted of cats previously diagnosed with CKD that were recruited through the research facility of the Small Animal Clinic at the Faculty of Veterinary Medicine between September 2021 and February 2024.

Eligibility for inclusion was determined at baseline using a standardized questionnaire (designed to assess current health status, dietary and medical treatment history), physical examination, systolic blood pressure (SBP) measurement (Doppler method), CBC, serum biochemistry profile (including symmetric dimethylarginine [SDMA], total thyroxine and electrolytes), feline leukemia virus (FeLV) antigen testing, feline immunodeficiency virus (FIV) serology, and complete urinalysis including urinary protein-to-creatinine (UPC) and bacterial culture, adhering to established protocols [20–24]. The key criteria for CKD diagnosis were serum creatinine concentration (sCr) > 1.83 mg/dL (> 161.8 μmol/L), urinary specific gravity (USG) < 1.035, and compatible history. The sCr cut-off in our study was based on a previously established laboratory-specific reference interval comprising 130 healthy cats [25]. Cats showing evidence of any other clinically relevant non-CKD related health abnormalities, based on the questionnaire or on the listed examinations, were excluded. Cats receiving treatments unrelated to CKD (e.g., non-steroidal anti-inflammatory drugs, antibiotics) were not included. Cats already receiving anti-emetics, gastroprotective medications, or appetite stimulants were excluded from the study. Cats with stage 1 or stage 4 CKD at baseline, as defined by International Renal Interest Society (IRIS), were excluded [26].

The cats were followed for 1 year at 6-month (i.e., T0, T6, and T12) intervals using the same tests (except for FIV/FeLV screening) and laboratory. Development of rapid CKD progression was the outcome of interest. Cats that developed other non-CKD related comorbidities were excluded. Therapeutic management of the included cats followed the IRIS guidelines [26]. Prescribed diets and medication were recorded (e.g., amlodipine, telmisartan, benazepril, phosphate binders). All cats were fasted before sampling at all timepoints.

The study population comprised:

I. A rapid disease progression cohort including all cats from the eligible population that experienced rapid CKD progression. Given the absence of a clear definition of rapidly progressive CKD in cats, progression criteria for our study were modified from those proposed previously [10]. Therefore, rapid progression was defined as a 25% increase from baseline sCr, occurring either before or at the 6-month (i.e., T6) follow-up interval. Re-examinations were offered before the 6-month time point if concerns about kidney deterioration arose.

II. A non-progressive cohort including all cats from the eligible population that demonstrated stable disease over 1 year. Cats were classified into the non-progressive group if their sCr did not exceed a 25% increment from baseline at the end of the 12-month follow-up period (i.e., T12).

At every visit (i.e., T0, T6, and T12) blood samples were acquired by venipuncture, specifically from the jugular or cephalic veins (23-gauge needle) and collected in plain serum tubes (VACUETTE Serum Tubes with Gel). The samples were allowed to clot for a maximum of 15 min at room temperature, followed by centrifugation (5 min; 1931×g) at 4°C. Within a narrow time window (15 min), urine samples were obtained by ultrasound-guided cystocentesis, utilizing a 22-gauge needle and transferred to plain tubes. The samples then were centrifuged (3 min; 355×g) at room temperature. After centrifugation, both the serum and urine supernatants were aliquoted into plastic Eppendorf vials and preserved at a temperature of −80°C until further analysis.

Baseline (T0) samples were used from identified cats for targeted metabolomics, specifically analyzing tryptophan and its metabolites (i.e., kynurenine, serotonin, and indole pathways), a panel of gut-derived uremic toxins (Table S1) and their precursors with the goal of comparing rapidly progressive and non-progressive cohorts before development of progression. Additionally, we evaluated the predictive capacity of specific metabolites and metabolite ratios for identifying rapidly progressive CKD.

2.3 | Metabolomic Profiling

All analytical standards (metabolites of interest; Table S1) and internal standards (ISTDs) were purchased from reputable sources (Sigma-Aldrich, St. Louis, Missouri, USA; ICN Biomedicals Inc., Ohio, USA). Ultrapure water and acetonitrile, both acidified using 0.1% formic acid, were used as analytic solvents. All solvents utilized in this experiment were of liquid chromatography—mass spectrometry (LC–MS) grade for extraction purposes and sourced from Fisher Scientific (Loughborough, UK) and VWR International (Merck, Darmstadt, Germany). Ultrapure water was obtained using a purified water system (Arium 611UV, Sartorius, Germany).

We designed, optimized, and validated specific extraction protocols for feline serum and urine samples targeting selected metabolic pathways (Table S1) [27].

For both biomatrices, the same ultra-high performance liquid chromatography (UHPLC) and mass spectrometry (MS) settings were applied. For chromatographic separation, a Vanquish Horizon UHPLC system (Thermo Fisher Scientific, San José, CA, USA) was used with an Acquity UPLC HSS C18 column T3 1.8 μm (150 mm × 2.1 mm; Waters, Manchester, UK) at a column temperature of 45°C. An injection volume of 10 μL at a maximum injection time of 150 ms was applied. A constant flow rate of 0.4 mL/min was maintained throughout the analysis.

An Exploris 120 stand-alone bench top orbitrap high-resolution mass spectrometer (Thermo Fisher Scientific, San José, CA, USA), equipped with a heated electrospray ionization source (HESI II), operating in polarity switching full scan mode at a scan range of 53–800 Da and with a mass resolution of 120 000 full width at half maximum (1 Hz) was used. Before mass spectrometric detection, initial instrument calibration was achieved by infusing ready-to-use calibration mixtures (Thermo Fisher Scientific, San José, CA, USA).

For each biomatrix, quality control samples (QCs) were generated by pooling sample extracts specific to the respective matrix. To adjust for instrumental fluctuations, QCs were run in duplicate after every sequence of 10 samples. Furthermore, at the start of the analyses, samples from the QC pool were injected to facilitate column conditioning, tailored to the specific biomatrix under investigation. An analytical standard mixture containing all a priori selected analytical standards (i.e., metabolites of interest, $n = 20$) was injected before, in the middle, and at the end of the analyses (Table S1). This procedure enabled the assessment of the operational parameters of the device and ensured accurate annotation of the designated metabolites.

2.4 | Data Analysis

Targeted data processing was executed using the Xcalibur 3.0 software (Thermo Fisher Scientific, San José, CA, USA). In this process, the identification of compounds relied on their m/z-value, carbon-isotope (C-isotope) profile, MS/MS fragmentation spectra, and retention time in relation to that of the internal standard [27]. Compound abundance was determined based on the total area under the curve (AUC). Subsequently, the data underwent QC normalization.

We performed statistical analysis using R (version 4.2.1) and SPSS Statistics (version 29) as follows:

a. Descriptive analyses were performed to summarize the data for the two cohorts. These included baseline characteristics, clinicopathological data, and metabolite abundance between the two groups.

b. Comparisons of serum and urine abundance of specific metabolites, along with routine clinical variables, were made between the rapidly progressive and non-progressive cohorts at baseline (before the onset of rapid progression; T0). Routine variables also were compared at the T6 timepoint (i.e., at detection of rapid progression). For this, the non-parametric independent Mann–Whitney U-test (for non-normally distributed data) or Student's t-test was employed (for normally distributed data). Normality was evaluated using the Shapiro–Wilk test, and the data distribution was visualized using histograms and QQ-plots. The Fisher exact test was used for comparison of proportions. To address the issue of multiple comparisons, a post hoc correction procedure based on false discovery rates (FDR), specifically the Benjamini-Hochberg method, was applied. Two group comparisons were visualized by boxplots.

c. Subsequently, metabolites and related metabolite ratios showing discriminative power around the significance threshold (p value < 0.05) were selected to assess associations with the

onset of rapid CKD progression. Binary logistic regression models were developed to quantify the association between each of the analyzed metabolites and rapid CKD progression, controlling for the confounding effect of sCr as a proxy measure for the stage of CKD at the time of study enrollment.

d. Lastly, the sensitivity and specificity of the selected metabolites in predicting rapid CKD progression were evaluated using receiver operating characteristic (ROC) curve analysis. The optimal thresholds were determined by applying Youden's index. Post-test probabilities were calculated using Bayes' theorem, incorporating the pre-test prevalence within the cohort to evaluate the predictive utility of compounds for rapid CKD progression in cats.

3 | Results

3.1 | Population Characteristics and Routine Variables

A detailed cohort follow-up and outcome summary is provided in the flow diagram in Figure 1. Of the eight cats in the rapidly progressive CKD cohort, three were screened before the 6-month timepoint because of rapid clinical deterioration. The rapidly progressive CKD cohort ($n=8$) consisted of three spayed females and five neutered males, whereas the non-progressive cohort ($n=34$) included 21 spayed females and 13 neutered males. No differences were observed in the age distribution between cats with rapidly progressive CKD and those with non-progressive CKD (Table 1). No significant difference was found in the proportion of cats being fed 100% renal diet (at time of sample collection) between the rapid progression (4/8; 50%) and non-progressive (18/34; 52.9%) cohort ($p=0.84$). However, shortly after the first visit, four of these 34 cats were newly transitioned from a regular diet to a renal diet ultimately leading to 22/34 cats being fed 100% renal diet. Furthermore, one of eight cats with rapidly progressive CKD first were switched from a renal diet back to a regular diet shortly after the visit because of reluctance of the cat to eat the renal diet. As a result, only 3/8 cats in this group were exclusively fed a renal diet. Therefore, significantly more cats in the non-progressive group (64.7%) were fed a 100% renal diet between baseline and 6 months of follow-up compared with the rapid progression group (37.5%; $p<0.01$). Additionally, groups did not differ with respect to prescriptions with renin angiotensin aldosterone system (RAAS)-inhibiting medication (Table 1).

The rapid progression cohort included 3 cats with IRIS stage 2 CKD and 5 with IRIS stage 3 CKD. In the non-progressive group, 23 cats had IRIS stage 2 and 11 had IRIS stage 3 CKD. In cats experiencing rapidly progressive CKD, baseline serum urea and phosphate concentrations and UPC were significantly higher (Table 1 and Figure 2). At baseline, no significant differences were found between the two groups in sCr, SDMA, hematocrit, USG, systolic blood pressure, body weight and body or muscle condition score. However, by the T6 timepoint, sCr, SDMA, urea, phosphate, and UPC were significantly higher in the rapid progression group, whereas USG and hematocrit were significantly lower.

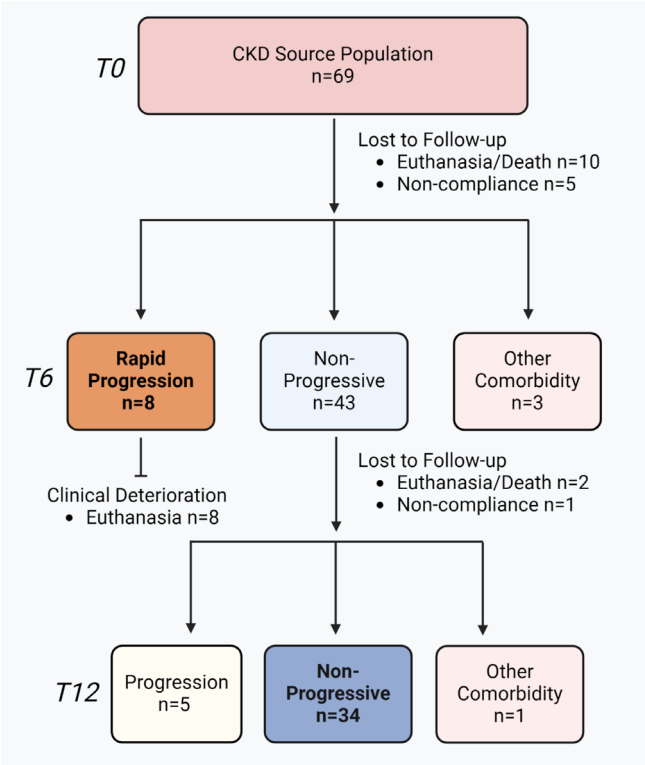

FIGURE 1 | Longitudinal flow diagram. The study population was divided into two cohorts: (1) A rapid progression cohort: Rapid progression was defined as >25% increase in serum creatinine over 6 months in conjunction with clinical deterioration. (2) A non-progressive control cohort: The control group had <25% increase in serum creatinine over 12 months. Cats developing comorbidities or experiencing moderate progression (>25% increase over 12 months) were excluded.

3.2 | Serum Metabolites and Rapid CKD Progression

Cats with rapidly progressive CKD had significantly lower abundance of serum L-tryptophan at baseline than cats with non-progressive CKD (mean ± SD, 0.69 ± 0.14 vs. 0.89 ± 0.20; Cohen's $d=1.03$, 95% CI, 0.22–1.82; $t(40)=2.612$; $p=0.02$). Moreover, tryptophan metabolism was further associated with CKD progression because metabolites of the serotonin pathway 5-hydroxytryptophan (median [IQR], 1.94 [1.55–2.59] vs. 1.12 [0.81–1.71]; $r=0.42$; $Z=2.65$; $p=0.01$) and 5-hydroxyindole-3-acetic acid (median [IQR], 1.60 [1.24–2.21] vs. 1.08 [0.86–1.46]; $r=0.36$; $Z=2.3$; $p=0.03$) were significantly higher at baseline in the serum of cats with rapidly progressive CKD. Serotonin itself, however, was not significantly altered (median [IQR], 0.87 [0.56–1.06] vs. 0.99 [0.74–1.18]; $r=0.21$; $Z=-1.38$; $p=0.18$; Figure 3). Notably, no other significant changes were observed in serum metabolites of interest, regarding the kynurenine pathway or gut-derived uremic toxins (Table 2).

To evaluate the predictive power of serotonin pathway metabolites in forecasting rapid progression of CKD in cats, we employed univariate binary logistic regression analysis (Table 3). Chi-square characteristics (L-tryptophan: $\chi^2=7.09$, $p<0.01$; 5-hydroxytryptophan: $\chi^2=6.11$, $p=0.01$; 5-hydroxyindole-3-acetic:

TABLE 1 | Baseline (T0) characteristic and comparison of routine parameters between rapid progression and non-progressive feline chronic kidney disease cohorts before progression onset.

Variable	T0	Rapid progression	Non-progressive	*p*	T6	Rapid progression	Non-progressive	*p*
Age (years)		Median: 12 IQR: [9–16] $n = 8$	Median: 13 IQR: [9–16] $n = 34$	0.72				
Renal diet (100%)[a]		36.5% $n = 3/8$	64.7% $n = 22/34$	< 0.01				
RAAS-inhibitors[b]		25% $n = 2/8$	44% $n = 15/34$	0.44				
Body weight		Mean: 4.05 STD: 0.93 $n = 7$	Mean: 4.12 STD: 1.01 $n = 32$	0.87				
Body condition score		Median: 3 IQR: [4, 5] $n = 8$	Median: 3 IQR: [4, 5] $n = 34$	0.83				
Muscle condition score		Median: 2 IQR: [2, 3] $n = 8$	Median: 2 IQR: [2] $n = 34$	0.34				
Creatinine (μmol/L)		Median: 277.6 IQR: [208.2–327.9] $n = 8$	Median: 219 IQR: [195.8–270.5] $n = 34$	0.13		Median: 455 IQR: [370.1–513.5] $n = 8$	Median: 200.7 IQR: [178.6–251.2] $n = 34$	< 0.01
Urea (mmol/L)		Median: 20.5 IQR: [15.2–31.5] $n = 8$	Median: 15.3 IQR: [13.5–17.6] $n = 34$	< 0.01		Median: 36 IQR: [29.8–45.2] $n = 8$	Median: 15 IQR: [12.7–17.4] $n = 34$	< 0.01
SDMA (μg/dL)		Median: 22.1 IQR: [16.0–36.7] $n = 8$	Median: 18 IQR: [14.6–25.4] $n = 34$	0.16		Median: 35.4 IQR: [29.2–46] $n = 8$	Median: 18.1 IQR: [14–20.9] $n = 34$	< 0.01
Phosphate (mmol/L)		Median: 1.47 IQR: [1.41–1.81] $n = 8$	Median: 1.30 IQR: [1.15–1.43] $n = 34$	0.03		Median: 2.25 IQR: [1.77–3.83] $n = 8$	Median: 1.42 IQR: [1.29–1.6] $n = 34$	< 0.01
Hematocrit (%)		Mean: 30 STD: 4.7 $n = 8$	Mean: 34.4 STD: 6.1 $n = 32$	0.07		Mean: 26.5 STD: 5 $n = 8$	Mean: 35.5 STD: 5.9 $n = 33$	< 0.01

(Continues)

TABLE 1 | (Continued)

Variable	T0	Rapid progression	Non-progressive	*p*	T6	Rapid progression	Non-progressive	*p*
Urinary specific gravity (USG)		Median: 1.018	Median: 1.020	0.45		Median: 1.015	Median: 1.021	< 0.01
		IQR: [1.015–1.025]	IQR: [1.016–1.030]			IQR: [1.012–1.017]	IQR: [1.016–1.029]	
		$n = 8$	$n = 34$			$n = 7$	$n = 33$	
Urinary protein-to-creatinine ratio (UPC)		Median: 0.39	Median: 0.20	0.02		Median: 0.5	Median: 0.15	< 0.01
		IQR: [0.29–0.42]	IQR: [0.13–0.31]			IQR: [0.48–2.4]	IQR: [0.11–0.33]	
		$n = 8$	$n = 34$			$n = 7$	$n = 33$	
Systolic blood pressure (mmHg)		Mean: 164	Mean: 155	0.29		Insufficient data		
		STD: 12	STD: 20					
		$n = 7$	$n = 30$					

Note: Comparisons at the time of rapid progression onset (T6).
[a]Percentage of cats fed exclusively a 100% renal diet relative to the total number in each cohort.
[b]Percentage of cats on RAAS inhibitors (benazepril or telmisartan) relative to the total number in each cohort.

$\chi^2 = 8.37$, $p < 0.01$) indicated that the tested metabolites were associated with rapidly progressive CKD. Furthermore, the pseudo-R^2 values of the models suggest that variations in L-tryptophan, 5-hydroxytryptophan, and 5-hydroxyindole-3-acetic accounted for approximately 25.0%, 23.5%, and 29.0% of the variance, respectively, in the progression of CKD in this cohort. Unit increases in L-tryptophan were associated with a nearly 100% decrease in the odds of developing CKD (OR, <0.01; 95% CI, 0.0–0.6; $p = 0.03$). In contrast, unit increases in 5-hydroxytryptophan and 5-hydroxyindole-3-acetic acid were associated with 2.8-fold (95% CI, 1.2–7.0; $p = 0.03$) and 10.3-fold increases (95% CI, 1.2–90; $p = 0.04$) in the odds of developing CKD, respectively. However, the association with 5-hydroxytryptophan became insignificant after adjusting for sCr ($p = 0.09$). Notably, the ratio of tryptophan to 5-hydroxytryptophan (OR, 0.01; 95% CI, 0.00–0.33; $p = 0.02$) and serotonin to its catabolite 5-hydroxyindole-3-acetic acid (OR, 0.01; 95% CI, 0.00–0.55; $p = 0.02$) were strongly associated with rapidly progressive CKD, maintaining significance after correcting for sCr (Table 3).

The ROC curve analysis further identified that L-tryptophan and 5-hydroxytryptophan have potential as predictive biomarkers for rapid disease progression in our feline CKD cohort with an area under the curve (AUC) of 0.76 (95% CI, 0.60–0.92; $p = 0.02$) and 0.82 (95% CI, 0.69–0.96; $p < 0.01$), indicating good discriminatory power translating into a maximum sensitivity and specificity of 75% and 61.8%, respectively, for L-tryptophan and 85.7% and 69.7%, respectively, for 5-hydroxytryptophan (Figure 4). Interestingly, the ratio of the latter compounds even demonstrated superior predictive ability with an AUC of 0.89 (95% CI, 0.78–1.0; $p < 0.01$) and a sensitivity and specificity of 85.7% and 78.8%, respectively, at the optimal threshold. Lastly, the ratio of serotonin to its catabolite, 5-hydroxyindole-3-acetic acid, also showed promising potential for predicting rapid CKD progression in cats (AUC, 0.83; 95% CI, 0.64–1.0; $p < 0.01$; sensitivity, 87.5%; specificity, 76.5%), surpassing the individual predictive ability of 5-hydroxyindole-3-acetic acid (AUC, 0.76; 95% CI, 0.53–0.99; $p < 0.01$; sensitivity, 87.5%; specificity, 66.6%; Figure 4).

Using Bayes' Theorem, we provide an example of how L-tryptophan and 5-hydroxytryptophan testing might be used in clinical practice. When presented with a cat with CKD, a clinician might make the plausible assumption that the (pre-test) probability of the cat being a rapid progression case is in the order of 20% (based on 8 rapid progression CKD cases among the 42 CKD cases in our study). If a positive result to an L-tryptophan test is returned, the revised (i.e., post-test) probability of this cat being a rapid progression CKD case is 32%. Similarly, if the same cat has a positive result on a 5-hydroxytryptophan test, the post-test probability of being a rapid progression CKD case is 40%.

3.3 | Urine Metabolites and Rapid CKD Progression

Cats with rapidly progressive CKD had significantly lower urinary abundance of indoxyl-sulfate at baseline than cats with non-progressive CKD (median [IQR], 0.25 [0.07–0.46] vs. 0.53 [0.36–0.81]; $r = 0.0.36$; $Z = -2.31$; $p = 0.04$; Table 4 and Figure 5). Moreover, the indole pathway was further associated with CKD progression because urine indole-3-acetic acid (median [IQR], 0.04 [0.03–0.15] vs. 0.27 [0.08–0.83]; $r = 0.39$; $Z = -2.53$; $p = 0.04$) was significantly lower at baseline in cats with rapidly progressive CKD. Furthermore, the excretion of the uremic toxin p-cresyl-sulfate was lower in the rapidly progressive CKD group (median [IQR], 0.13 [0.01–0.79] vs. 0.83 [0.32–1.27]; $r = 0.37$; $Z = -2.4$; $p = 0.04$). Another gut-derived uremic toxin, urinary trimethylamine-N-oxide (TMAO), was not significantly lower in cats with rapidly progressive CKD compared to those with non-progressive CKD (median [IQR], 0.38 [0.20–0.64] vs. 0.60 [0.43–1.14]; $r = 0.27$; $Z = -1.77$; $p = 0.40$; Figure 5). No other significant changes were observed

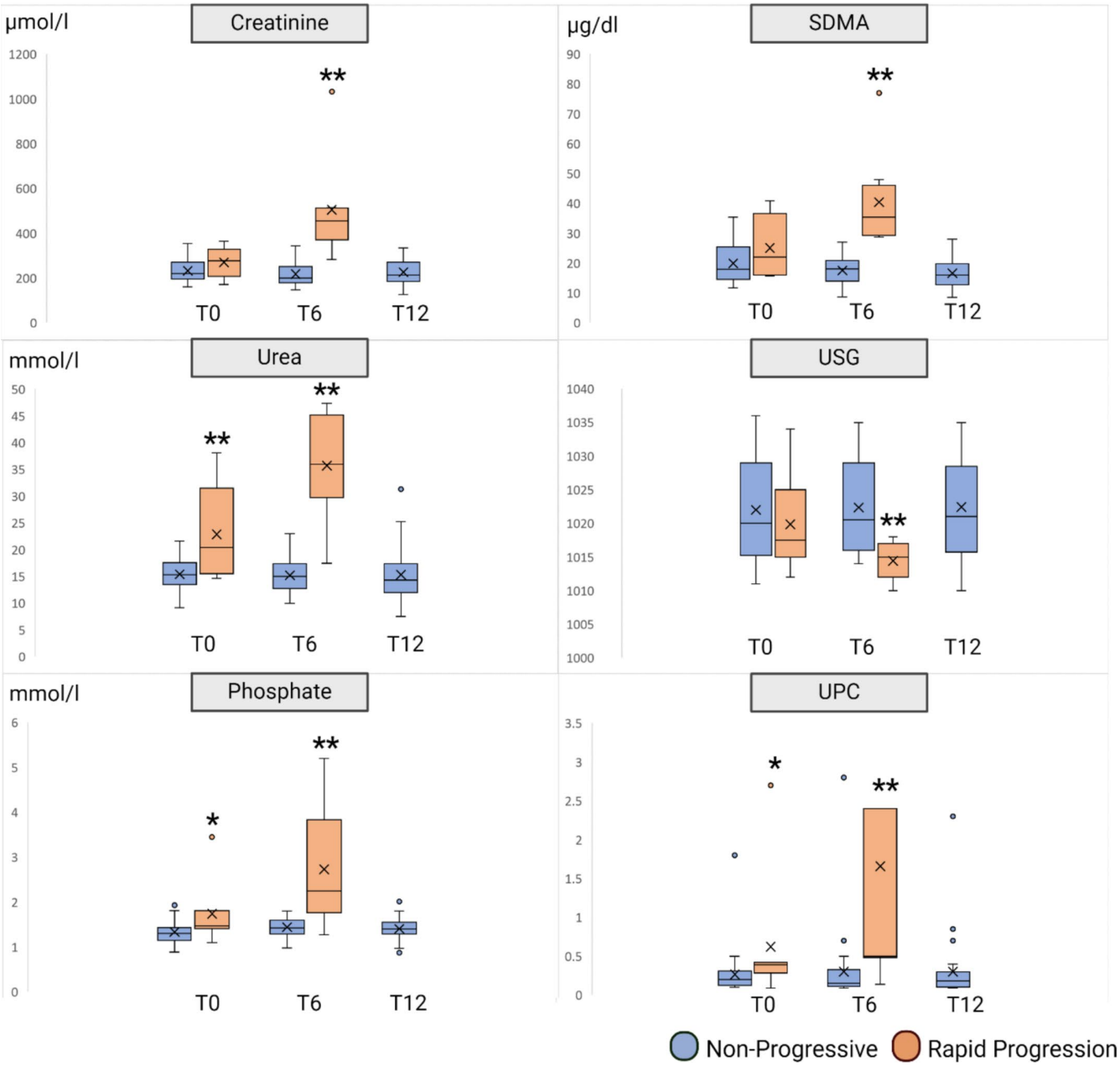

FIGURE 2 | Box-plot representation illustrating a two-group comparison of routine renal parameters between feline patients with chronic kidney disease, categorized into a rapidly progressive ($n=8$) and a non-progressive cohort ($n=34$) at baseline (T0) and after 6 months (T6) of follow-up (p value $<0.05=*$; $<0.01=**$). A third timepoint (after 12 months of follow-up) is provided for the non-progressive cohort.

in urine metabolites of interest (Table 4). To further evaluate the effect of renal excretion, serum-to-urine ratios of the listed gut-derived uremic toxins were compared between groups. The serum-to-urine ratio of indoxyl-sulfate was significantly higher in the rapid CKD progression group (median [IQR], 3.08 [2.05–8.11] vs. 1.86 [1.44–2.58]; $r=0.34$; $Z=2.21$; $p=0.03$). However, no significant differences were found between groups for the serum-to-urine ratios of p-cresyl-sulfate (median [IQR], 2.62 [1.43–7.43] vs. 2.01 [1.49–7.43]; $r=0.20$; $Z=1.3$; $p=0.20$) and TMAO (median [IQR], 1.61 [0.95–3.18] vs. 1.58 [1.16–2.25]; $r=0.01$; $Z=0.06$; $p=0.97$).

Chi-square statistics indicated that urinary indoxyl-sulfate, p-cresyl-sulfate, and TMAO were individually associated with rapidly progressive CKD (indoxyl-sulfate: $\chi^2=5.76$, $p=0.02$; p-cresyl-sulfate: $\chi^2=5.10$, $p=0.02$; TMAO: $\chi^2=4.95$, $p=0.03$). Indole-3-acetic acid ($\chi^2=0.44$, $p=0.51$) however, did not show an association with disease progression.

Furthermore, the pseudo-R^2 values indicated that urinary variations in indoxyl-sulfate, p-cresyl-sulfate, and TMAO explained approximately 20.6%, 18.4%, and 17.8% of the variance, respectively, in the progression of CKD in our cohort of cats.

The logistic regression analyses identified urinary indoxyl-sulfate as a potential marker associated with rapidly progressive CKD (OR, 0.03; 95% CI, 0.00–1.0; $p=0.05$). Accounting for sCr attenuated the association (Table 5). Urinary p-Cresyl-sulfate

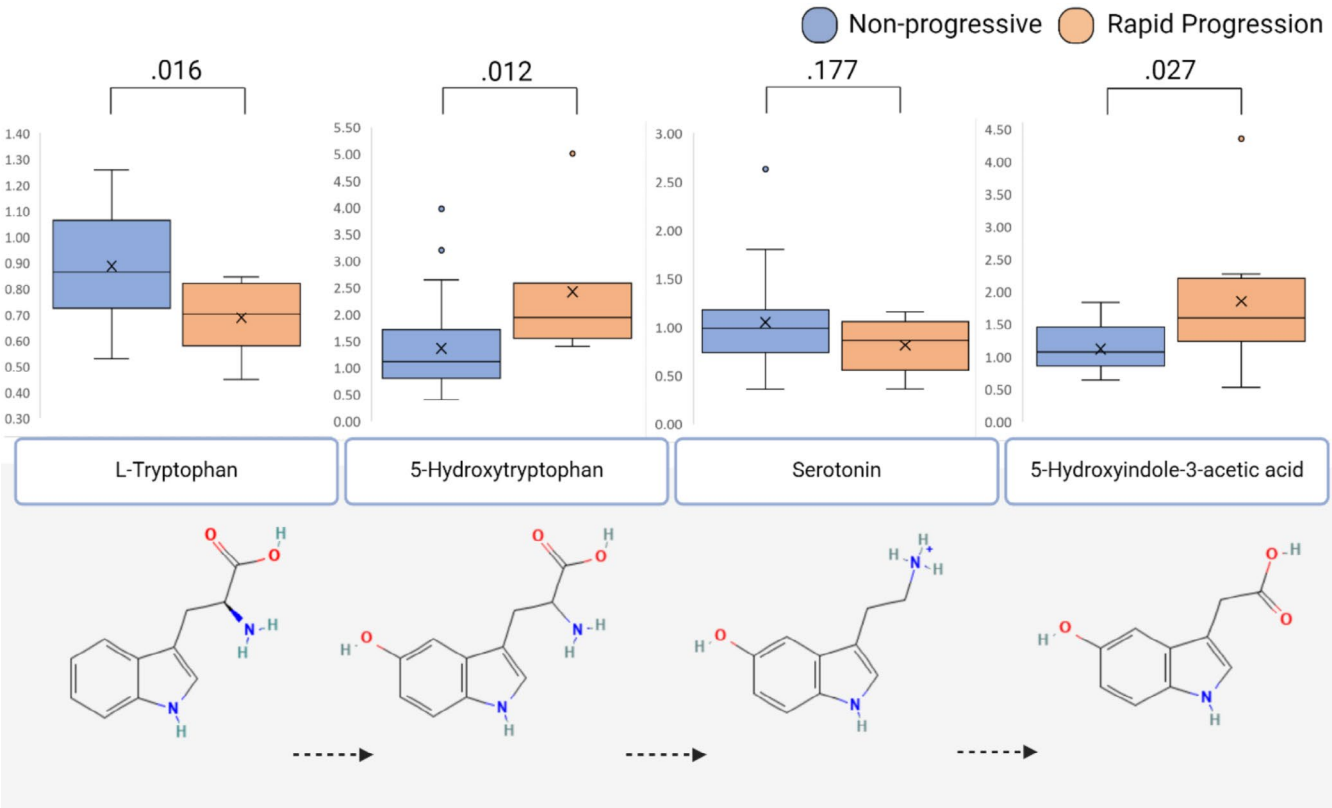

FIGURE 3 | Box-plot representation illustrating a two-group comparison (relative abundance) of serum metabolites from the serotonin pathway between feline patients with chronic kidney disease, categorized into a rapidly progressive ($n = 8$) and a non-progressive cohort ($n = 34$). p values are represented after FDR correction.

(OR, 0.15; 95% CI, 0.02–1.1; $p = 0.06$) and TMAO (OR, 0.07; 95% CI, 0.01–1.2; $p = 0.07$), demonstrated potential as predictors of rapid disease progression, but their predictive strength was decreased after adjustment for sCr (Table 5).

The ROC curve analysis demonstrated the potential of urinary indoxyl-sulfate and p-cresyl-sulfate as predictive biomarkers for rapid disease progression in our feline CKD cohort with an AUC of 0.77 (95% CI, 0.59–0.94; $p < 0.01$) and 0.78 (95% CI, 0.57–0.98; $p = 0.01$) and an equally moderate sensitivity for both compounds of 75% and a specificity of respectively 67.6% and 73.5% at the optimal threshold (Figure 6). The characteristics (sensitivity, 62.5%; specificity, 64.7%) for TMAO when predicting rapidly progressive CKD were found to be suboptimal (AUC, 0.70; 95% CI, 0.51–0.89; $p = 0.04$).

If a positive result for a urinary indoxyl-sulfate, p-cresyl-sulfate, and TMAO test is returned, the revised (i.e., post-test) probability of the cat being a rapid progression CKD case is, respectively, 32%, 36%, or 29%.

4 | Discussion

Our study aimed to explore compounds of tryptophan metabolism, together with key gut-derived toxins associated with rapidly progressive CKD in cats in a prospective cohort study with a comparative analysis of predictors using a targeted metabolomic approach.

The primary finding was that tryptophan metabolism was significantly upregulated along the serotonin pathway in cats with rapidly progressive CKD. Tryptophan can only be acquired from dietary sources and is metabolized through the kynurenine, indole, and serotonin pathway [28]. Interestingly, increased formation of the serotonin precursor, 5-hydroxytryptophan from tryptophan, was associated with disease progression whereas tryptophan itself was negatively associated with progressive CKD. In contrast, serum serotonin remained unchanged, whereas its catabolite, 5-hydroxyindole-3-acetic acid, was associated with rapidly declining kidney function in these cats. Furthermore, the serum L-tryptophan/5-hydroxytryptophan and serotonin/5-hydroxyindole-3-acetic acid ratios demonstrated promising predictive ability for rapid disease progression in this cohort. The study estimated a 20% pre-test probability of rapid CKD progression in cats. Biomarkers related to tryptophan metabolism enhanced predictive guidance. For instance, 5-hydroxytryptophan increased the post-test probability to 40%. Although these findings are promising, they require further validation because of the small sample size and the specialized methods employed. Nonetheless, the results emphasize the potential for improved risk stratification and clinical decision-making.

In the context of biochemical pathway analysis, 5-hydroxyindole-3-acetic acid formation results from enzymatic modulation by mitochondrial monoamine oxidase A (MAO-A), which is also expressed in renal tubular cells [29]. Upregulation of MAO-A is associated with pro-inflammatory

TABLE 2 | Comparisons of baseline (T0) serum metabolite abundance between a rapid chronic kidney disease progression cohort ($n = 8$) and a non-progressive cohort ($n = 34$) before progression onset.

Variable	Rapid progression ($n = 8$)	Non-progressive ($n = 34$)	p[a]
L-Tryptophan	Mean: 0.69	Mean: 0.89	0.02
	STD: 0.14	STD: 0.20	
Indole pathway			
Indole	Median: 1.07	Median: 0.96	0.47
	IQR: [0.72–2.46]	IQR: [0.50–2.16]	
Indoxyl-sulfate	Median: 0.83	Median: 1.10	0.72
	IQR: [0.27–1.59]	IQR: [0.67–1.61]	
Indole-3-acetic acid	Median: 0.86	Median: 0.64	0.96
	IQR: [0.48–1.00]	IQR: [0.44–1.88]	
Kynurenine pathway			
Kynurenine	Mean: 0.93	Mean: 1.03	0.80
	STD: 0.53	STD: 0.26	
Kynurenic acid	Median: 1.38	Median: 1.17	0.70
	IQR: [0.94–2.50]	IQR: [0.83–1.52]	
3-Hydroxykynurenine	Median: 1.20	Median: 1.10	0.79
	IQR: [0.97–1.39]	IQR: [0.90–1.32]	
Serotonin pathway			
5-Hydroxytryptophan	Median: 1.94	Median: 1.12	0.01
	IQR: [1.55–2.59]	IQR: [0.81–1.71]	
Serotonin	Median: 0.87	Median: 0.99	0.18
	IQR: [0.56–1.06]	IQR: [0.74–1.18]	
5-Hydroxyindole-3-acetic acid	Median: 1.60	Median: 1.08	0.03
	IQR: [1.24–2.21]	IQR: [0.86–1.48]	
Carnitine metabolism			
L-Carnitine	Median: 1.16	Median: 1.02	0.89
	IQR: [0.80–1.73]	IQR: [0.82–2.00]	
Betaine	Median: 0.81	Median: 0.63	0.56
	IQR: [0.56–0.91]	IQR: [0.48–0.96]	
Trimethylamine (TMA)	Median: 0.85	Median: 0.97	0.52
	IQR: [0.62–1.12]	IQR: [0.78–1.19]	
Trimethylamine-N-oxide (TMAO)	Mean: 0.86	Mean: 1.18	0.72
	STD: 0.67	STD: 0.50	
Tyrosine metabolism			
L-Tyrosine	Mean: 0.95	Mean: 0.96	0.71
	STD: 0.04	STD: 0.06	
p-Cresol	Median: 0.12	Median: 1.44	0.56
	IQR: [0.02–3.94]	IQR: [0.75–2.25]	
p-Cresyl-sulfate	Median: 0.12	Median: 1.50	0.28
	IQR: [0.02–3.78]	IQR: [0.76–2.19]	

Note: The metabolites are sorted according to their respective metabolic pathways.
[a] P value represented after FDR correction.

TABLE 3 | Details of the univariate binary logistic regression analysis for different serum metabolites and metabolite ratios from the serotonin pathway associated with onset of rapid chronic kidney disease (CKD) progression.

Covariate	Outcome positive subjects	Total subjects	Regression coefficient (SE)	Z-statistic	*p*	Odds ratio (95% CI)
L-Tryptophan	8	42	−6.450 (2.928)	4.6	0.03	0.00 (0.00–0.60)
5-Hydroxytryptophan	8	42	0.957 (0.561)	3.0	0.09	2.6 (0.88–7.8)
L-Tryptophan/5-Hydroxytryptophan	8	42	−8.044 (3.814)	4.5	0.03	0.00 (0.00–0.55)
5-Hydroxyindole-3-acetic acid	8	42	2.540 (1.241)	4.1	0.04	14.2 (1.1–182.3)
Serotonin/5-Hydroxyindole-3-acetic acid	8	42	−5.441 (2.555)	4.2	0.04	0.01 (0.00–0.83)

Note: Adjustment was made for serum creatinine to control for confounding effects related to disease stage. The odds of a 5-hydroxytryptophan positive cat being rapid progression CKD positive was 2.6 (95% CI 0.88–7.8) times that of a 5-hydroxytryptophan negative cat. This means that an increase in the serum abundance of this metabolite is strongly associated with a higher likelihood of rapid CKD progression. The wide confidence interval suggests variability in the estimate. Nevertheless, this highlights the potential importance of 5-hydroxytryptophan as a potential marker of CKD progression.

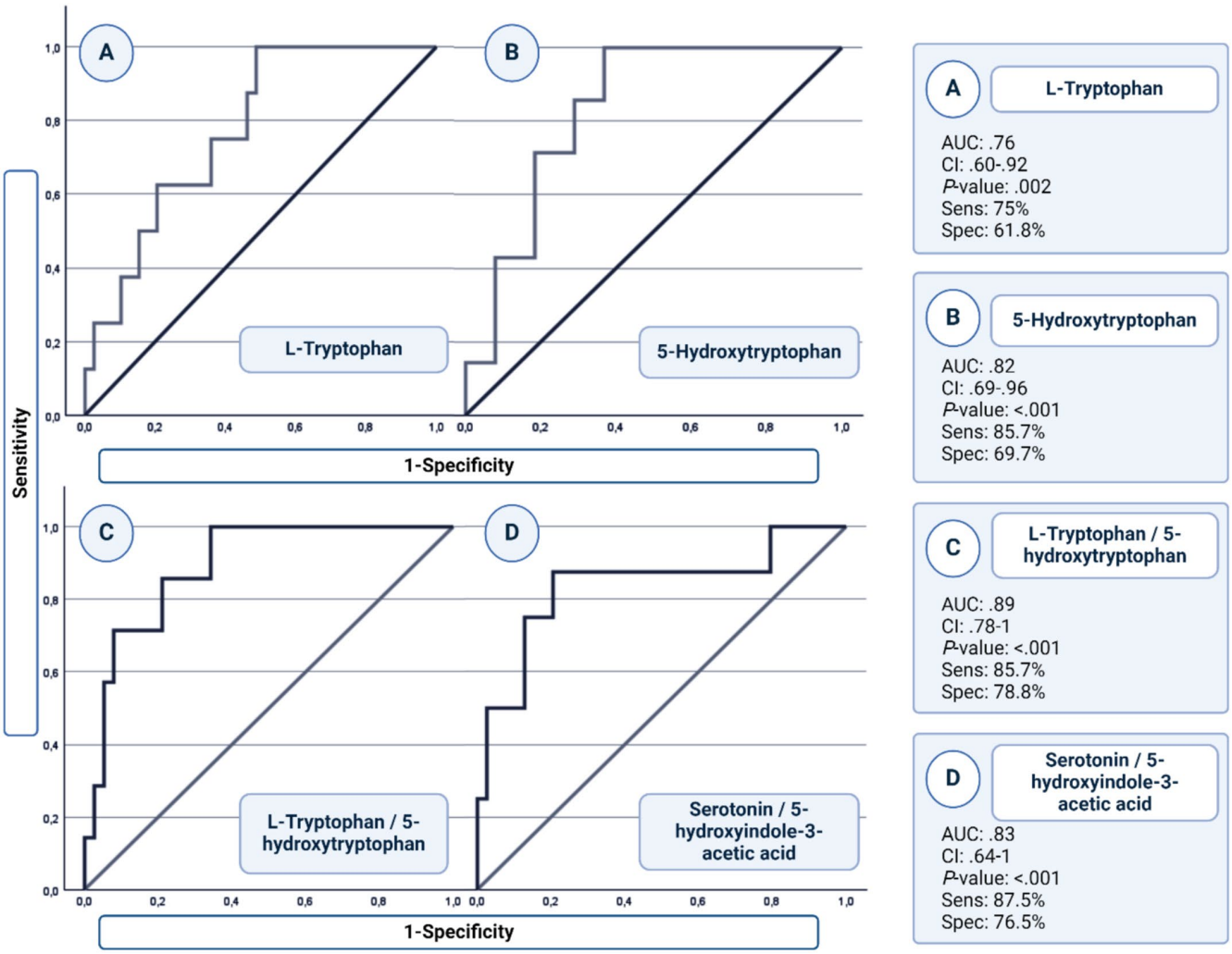

FIGURE 4 | Receiver operator characteristic (ROC) curves of metabolites and metabolite ratios of the serotonin pathway for assessment of their predictive ability for rapid disease progression in feline patients with chronic kidney disease. Rapid progression cohort ($n = 8$) and non-progressive cohort ($n = 34$).

conditions, and further aggravates free radical production in renal disease [30]. Therefore, the ratio of serotonin to 5-hydroxyindole-3-acetic acid can be considered a surrogate marker of MAO-A activity, indicative of ongoing oxidative stress, which may contribute to the acceleration of kidney dysfunction [31, 32]. Hence, 5-hydroxyindole-3-acetic acid is

TABLE 4 | Comparisons of baseline (T0) urine metabolite abundance between a rapid chronic kidney disease progression cohort ($n = 8$) and a non-progressive cohort ($n = 34$) before progression onset.

Variable	Rapid progression ($n = 8$)	Non-progressive ($n = 34$)	p[a]
L-Tryptophan	Median: 0.24	Median: 0.39	0.21
	IQR: [0.16–0.68]	IQR: [0.27–0.63]	
Indole pathway			
Indole	Median: 0.74	Median: 0.66	0.49
	IQR: [0.45–1.41]	IQR: [0.29–1.02]	
Indoxyl-sulfate	Median: 0.25	Median: 0.53	0.04
	IQR: [0.07–0.46]	IQR: [0.36–0.81]	
Indole-3-acetic acid	Median: 0.04	Median: 0.27	0.04
	IQR: [0.03–0.15]	IQR: [0.08–0.83]	
Kynurenine pathway			
Kynurenine	Median: 0.90	Median: 0.71	0.63
	IQR: [0.52–0.98]	IQR: [0.44–1.11]	
Kynurenic acid	Median: 0.53	Median: 0.78	0.74
	IQR: [0.30–1.15]	IQR: [0.40–1.12]	
3-Hydroxykynurenine	Median: 0.38	Median: 0.52	1
	IQR: [0.22–0.60]	IQR: [0.31–0.78]	
Serotonin pathway			
5-Hydroxytryptophan	Median: 1.30	Median: 1.03	0.86
	IQR: [0.74–2.36]	IQR: [0.68–1.50]	
Serotonin	Median: 0.40	Median: 0.34	0.78
	IQR: [0.56–1.06]	IQR: [0.18–0.84]	
5-Hydroxyindole-3-acetic acid	Median: 0.42	Median: 0.55	0.39
	IQR: [1.24–2.20]	IQR: [0.38–0.86]	
Carnitine metabolism			
L-Carnitine	Median: 0.53	Median: 0.48	0.84
	IQR: [0.20–0.87]	IQR: [0.20–0.87]	
Betaine	Median: 0.12	Median: 0.17	0.45
	IQR: [0.08–0.29]	IQR: [0.09–0.40]	
Trimethylamine (TMA)	Median: 0.28	Median: 0.47	0.32
	IQR: [0.19–0.43]	IQR: [0.25–0.76]	
Trimethylamine-N-oxide (TMAO)	Median: 0.38	Median: 0.60	0.40
	IQR: [0.20–0.64]	IQR: [0.43–1.14]	
Tyrosine metabolism			
L-Tyrosine	Median: 0.36	Median: 0.48	0.54
	IQR: [0.24–0.81]	IQR: [0.27–0.86]	
p-Cresol	Median: 0.31	Median: 0.83	0.19
	IQR: [0.02–1.03]	IQR: [0.32–1.33]	
p-Cresyl-sulfate	Median: 0.13	Median: 0.83	0.04
	IQR: [0.01–0.79]	IQR: [0.32–1.27]	

Note: The metabolites are sorted according to their respective metabolic pathways.
[a] P value represented after FDR correction.

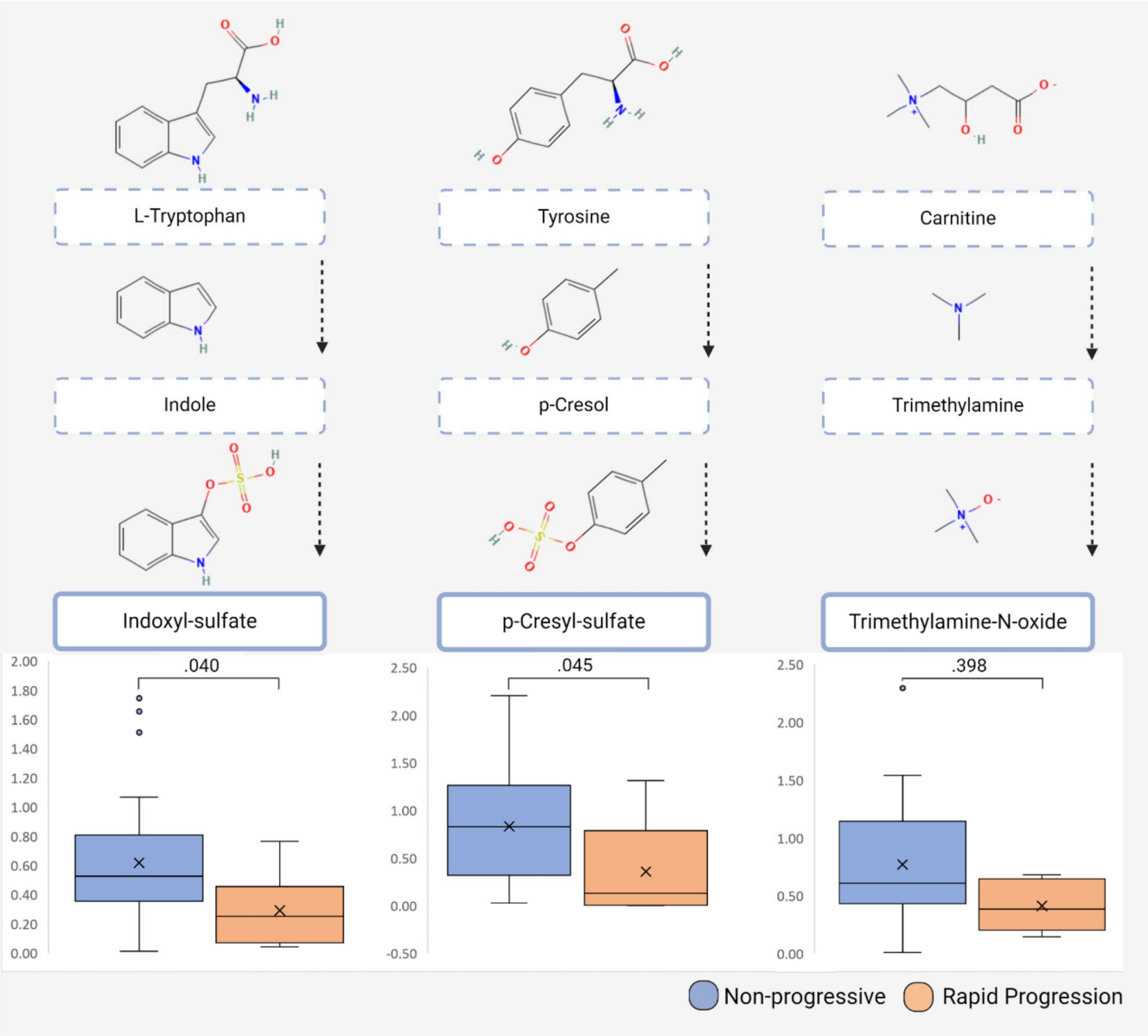

FIGURE 5 | Box-plot representation illustrating a two-group comparison (relative abundance) of urine metabolites from; (I) L-tryptophan catabolism (indole pathway); (II) tyrosine catabolism; (III) and carnitine catabolism, between feline patients with chronic kidney disease, categorized into a rapid progression ($n = 8$) and a non-progressive cohort ($n = 34$). *P*-value represented after FDR correction.

TABLE 5 | Details of the univariate binary logistic regression analysis for different urinary solutes, also classified as uremic toxins, associated with the onset of rapid chronic kidney disease (CKD) progression.

Covariate	Outcome positive subjects	Total subjects	Regression coefficient (SE)	Z-statistic	*p*	Odds ratio (95% CI)
Indoxyl-sulfate	8	42	−3.465 (1.743)	3.5	0.06	0.04 (0.01–1.2)
Indole-3-acetic acid	8	42	−0.341 (0.632)	0.30	0.58	0.70 (0.21–2.4)
p-Cresyl-sulfate	8	42	−1.970 (1.100)	3.4	0.07	0.15 (0.02–1.1)
Trimethylamine-N-oxide (TMAO)	8	42	−2.984 (1.552)	3.2	0.07	0.06 (0.00–1.3)

Note: Adjustment was made for serum creatinine to control for confounding effects related to disease stage. For Indoxyl-sulfate, the odds ratio is 0.04 (95% CI 0.01–1.2). This suggests that lower excretion of indoxyl-sulfate may be associated with a higher likelihood of rapid CKD progression. The lower limit of the confidence interval (0.01) indicates a potentially strong effect, though the upper limit (1.2) suggests uncertainty about the magnitude of the association. This finding warrants further investigation to confirm whether lower indoxyl-sulfate excretion might serve as a marker of CKD progression.

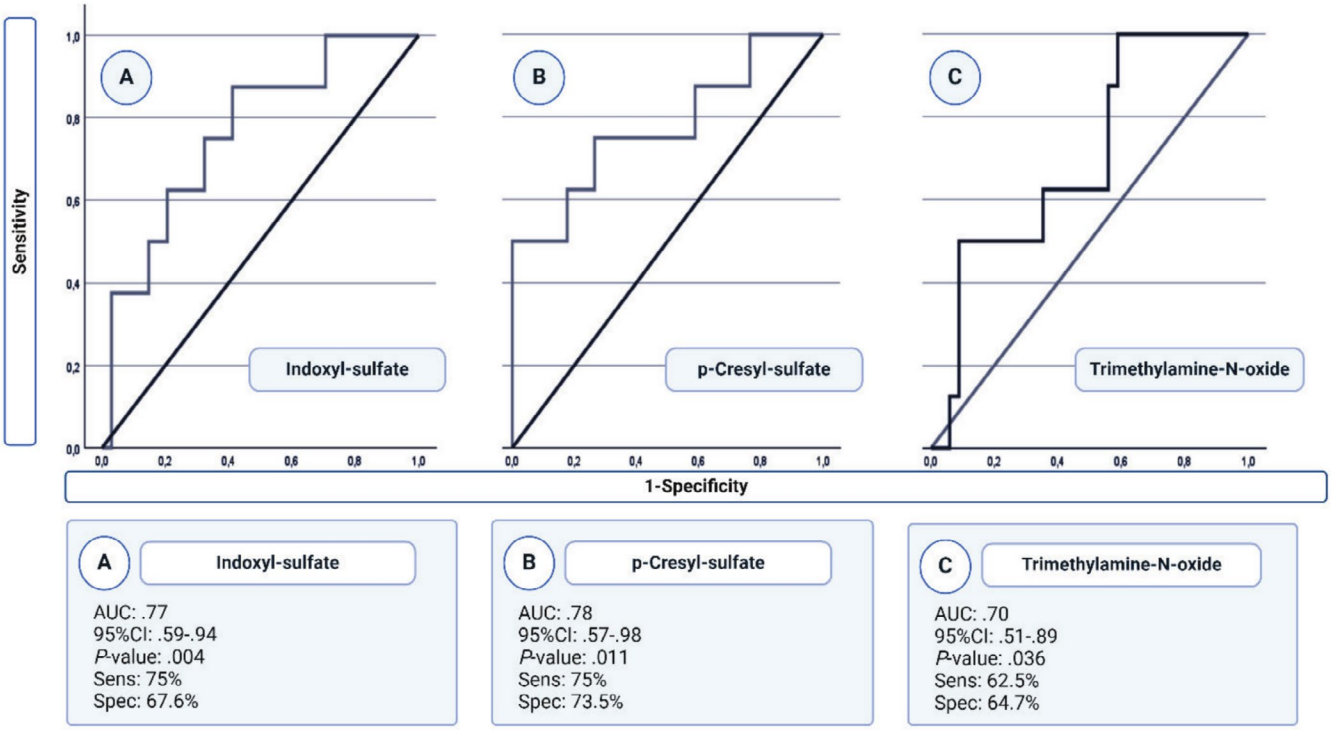

FIGURE 6 | Receiver operator characteristic (ROC) curves of different urinary solutes, also classified as uremic toxins, for assessment of their predictive ability for disease progression in feline patients with chronic kidney disease. Rapid progression cohort ($n = 8$) and non-progressive cohort ($n = 34$).

likely not a direct mediator of disease progression but rather may serve as an indicator of renal tubular inflammation. It is not clear if decreased circulating tryptophan was a direct result of an upregulated serotonin pathway in the progressive CKD cohort, because this pathway normally only accounts for 5% of tryptophan metabolism [28, 32]. In previous work, we showed that, along with upregulation of the serotonin pathway, metabolites of the kynurenine pathway (typically responsible for approximately 95% of dietary tryptophan degradation) also were upregulated in cats with CKD compared with healthy controls, likely contributing to decreased circulating tryptophan [15, 32]. However, in our study kynurenine metabolites were not associated with rapid CKD progression, suggesting that the decrease in circulating tryptophan may be driven primarily by upregulation of the serotonin pathway rather than the kynurenine pathway. In addition, we observed that the serum tryptophan/5-hydroxytryptophan ratio was associated with rapid CKD progression, which would support this hypothesis.

Furthermore, urinary abundance of the tryptophan-derived uremic toxin indoxyl-sulfate was significantly decreased in the rapidly progressive CKD cohort, whereas the serum-to-urine ratio was significantly higher. Indoxyl-sulfate formation from tryptophan occurs through an alternative pathway, distinct from the intrinsic kynurenine and serotonin pathways. In this process, gut bacterial enzymes degrade tryptophan into indole, which then is converted to indoxyl-sulfate in the liver [33]. Indoxyl-sulfate is then actively excreted by the renal tubular cells [33, 34]. It is well established that decreasing kidney function leads to higher circulating concentrations of indoxyl-sulfate. A number of different underlying mechanisms can explain this increase in cats with CKD: (I) Increased microbial formation and resorption of indole, along with abnormalities in the intestinal wall barrier and (II) decreased fractional renal excretion [15, 35]. Accumulation of this toxin in renal tubular cells may promote CKD progression by induction of oxidative stress, increased profibrotic cytokine production, and inhibition of tubular proliferation [36–39]. We were unable to validate an association between serum indoxyl-sulfate and rapid disease progression, in contrast to a previous study that highlighted indoxyl-sulfate as a predictor of disease progression in cats with CKD [40].

p-Cresyl-sulfate is another gut-derived uremic toxin formed through bacterial degradation of tyrosine and is retained in cats with CKD [15, 41]. p-Cresyl-sulfate contributes to many mechanisms involved in cell injury and fibrosis, including those occurring in the renal tubular epithelium [42]. Although, in human patients with CKD, serum p-cresyl-sulfate is considered a marker of all-cause mortality and cardiovascular complications, associations with progression of renal disease also have been established [43, 44]. Here, we could not confirm that serum p-cresyl-sulfate is a predictor of rapid disease progression. In contrast, urinary p-cresyl-sulfate excretion was lower in the rapid progression cohort but not significantly predictive of imminent kidney function deterioration. A third gut-derived uremic toxin, TMAO, is derived from dietary precursors betaine and carnitine, which are converted into the intermediate product trimethylamine (TMA) by gut bacterial enzymes [45]. Recently, a direct association between TMAO, tubulointerstitial fibrosis and progressive kidney dysfunction was observed in animal models [46, 47]. No clear association between serum TMAO and

CKD progression could be observed in our study. Furthermore, urinary excretion of TMAO did not significantly predict rapidly progressive CKD, with significance remaining near the threshold. The small number of cats in the rapid progression cohort may have limited the statistical power of the study, likely affecting these results.

The observed decrease in the urinary excretion of gut-derived uremic toxins, such as indoxyl-sulfate and p-cresyl-sulfate, in progressive CKD in cats may indicate ongoing tubular dysfunction or damage because these toxins primarily are excreted through the renal tubules [33, 34]. This hypothesis may be further supported by the observation that serum-to-urine ratios of indoxyl-sulfate were higher in the rapid progression group, suggesting decreased renal excretion in this group. These findings also may align with the fact that tubular lesions are strongly linked to subsequent kidney function deterioration [9]. Furthermore, the observation that cats of our rapidly progressive cohort had higher UPC values reinforces the hypothesis of underlying tubular dysfunction because mild proteinuria, such as observed in cats with CKD, is associated with inflammation or degeneration of the tubular compartment [9, 10]. Renal proteinuria is a well-recognized predictor of disease progression in cats, and decreasing dietary protein content may help in managing proteinuria as part of a broader therapeutic approach [10, 48]. However, to date, there is no proven evidence that medical treatment for dietary-unresponsive renal proteinuria leads to better long-term outcomes [49–51].

Interestingly, amino acid precursors of the studied uremic toxins are highly abundant in diets with a high protein content such as the strict carnivore diet of cats [48, 52]. Some studies suggest that decreasing protein in the diets of cats with CKD might help improve long-term outcomes [8, 53]. However, this hypothesis is still the subject of debate and lacks sufficient supporting data because renal diets have a complex composition, making it difficult to discern that the benefits are due solely to protein restriction. Conversely, providing enhanced concentrations of specific amino acids appears important to preserve lean body mass and may be effective in preventing CKD progression [54, 55]. Recent evidence suggests that dietary macronutrient composition can directly affect circulating concentrations of gut-derived uremic toxins in cats. For instance, a study in healthy cats found that those fed high-protein diets had significantly higher circulating indoxyl-sulfate concentrations [56]. However, a similar study in which cats were fed either a low-protein diet or a high-protein diet for 12 weeks did not identify changes in indoxyl-sulfate, p-cresyl-sulfate, or TMAO concentrations [57]. Findings from studies in healthy cats may not be directly applicable to cats with CKD. Indeed, in a study of cats with IRIS stage 1 CKD, higher dietary protein concentrations led to increased plasma concentrations of indoxyl-sulfate and p-cresol, as well as increased renal proteinuria (at 16 weeks), despite similar phosphorus content [58]. Furthermore, body weight and food intake remained unaffected across diets [58]. Currently, there is no clear understanding of the ideal amount of dietary protein for cats diagnosed with CKD, but it is generally advisable to avoid high-protein diets [7]. Although decreasing dietary protein theoretically may ease tubular stress by lowering the concentration of circulating uremic toxins, no long-term studies have directly demonstrated that these affect CKD progression in cats [4].

A key challenge in our study was the lack of a clear definition for rapid CKD progression in cats. Although sCr is widely used to assess disease stage, it can be influenced by non-renal factors (e.g., age, muscle mass, and dehydration), potentially over- or under-estimating progression [13, 14]. To specifically focus on rapid progression, we defined it as a 25% increase in baseline sCr after 6 months, rather than 12 months as in previous studies [6]. A limitation is the absence of a third timepoint to confirm progression, although 6-month data showed significantly increased indirect glomerular filtration (GFR) markers, decreased USG, and clinical deterioration, supporting true functional decline. However, distinguishing CKD progression from acute-on-chronic episodes (e.g., ureteral obstruction and pyelonephritis) was challenging without imaging. This factor represents a limitation in fully excluding these conditions. The possibility of misclassifying cats with ongoing progression in the non-progression group was mitigated by further follow-up of the latter cohort, which had to remain stable after 12 months of follow-up for inclusion. The relatively small number of cats in our study resulted in insufficient statistical power, making results preliminary and requiring further validation. Dietary effects on metabolites were not assessed, although baseline diet distribution was similar between groups, suspected to mitigate potential confounding. Additionally, metabolites were only measured once at baseline, despite known intra- and interindividual variability, which may have led to missed associations [59]. Finally, RAAS inhibitors and antihypertensives may have influenced metabolite concentrations.

In conclusion, metabolites of the serotonin pathway: L-tryptophan, 5-hydroxytryptophan, and 5-hydroxyindole-3-acetic acid may serve as candidates for future predictive validation in CKD progression in cats. Moreover, the decrease in gut-derived uremic toxin excretion in rapidly progressive CKD presents an initial hypothesis that tubular dysfunction, evidenced by impaired toxin handling, may underlie the rapid progression of the disease.

Acknowledgments

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

The study protocol was approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering, Ghent University (EC2020-081). Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.